

Global Public Health Indicators: A Data Mining Analysis of Health Determinants and Trends

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Data Mining Project Report

Abstract—Global Public Health Indicators provide essential metrics for evaluating population health status, guiding policy, and tracking progress toward Sustainable Development Goals (SDGs). This project utilizes data mining techniques to analyze multidimensional datasets covering life expectancy, mortality rates, immunization coverage, and socio-economic factors from 1980 to 2024. By employing correlation analysis, K-Means clustering, and Principal Component Analysis (PCA), we identify critical drivers of life expectancy, quantify the impact of vaccination on infant mortality, and map the epidemiological transition from infectious to chronic diseases. Key findings reveal that infant mortality ($r \approx -0.84$) and schooling ($r \approx 0.71$) are the strongest predictors of life expectancy. Furthermore, vaccination coverage accounts for approximately 65% of the variance in infant mortality rates. The study also highlights persistent health inequalities between income groups despite global improvements.

Index Terms—Data mining, Public Health, K-Means Clustering, Mortality Trends, Vaccination, PCA.

I. INTRODUCTION

Global health monitoring is critical for identifying disparities between developed and developing nations. Indicators such as life expectancy, infant mortality, immunization coverage, and disease burden serve as proxies for a nation's overall development and healthcare infrastructure stability.

The motivation for this study stems from the need to:

- Understand global population health dynamics.
- Identify inequalities between income groups.
- Measure progress toward global health goals (SDG-3).
- Evaluate the efficacy of interventions like vaccination programs.

This report aggregates data from the World Health Organization (WHO), UNICEF, and the World Bank to perform a granular analysis of these indicators. We focus specifically on the correlation between socio-economic factors and life expectancy, the measurable impact of measles and Hepatitis B vaccinations on survival rates, and the shifting burden of disease from infectious to chronic conditions.

II. METHODOLOGY

A. Dataset Description

The analysis incorporates multiple datasets spanning the years 1980 to 2024. Key features include:

- **Mortality Metrics:** Adult Mortality, Infant Mortality, Under-Five Deaths.

- **Socio-Economic Factors:** GDP, Schooling Years, Income Composition of Resources.
- **Immunization:** Hepatitis B (HepB), Measles, Polio, DTP coverage.
- **Disease Burden:** HIV/AIDS prevalence, DALYs (Disability-Adjusted Life Years), and cause-specific death counts (Cardiovascular, Malaria, TB).

B. Data Preprocessing

To ensure data quality and analytical reliability, the following preprocessing steps were undertaken:

- **Missing Value Imputation:** Numerical features with missing entries were filled using the median to mitigate the impact of outliers. Categorical variables were filled using the mode.
- **Duplicate Removal:** Duplicate records were identified and removed to maintain data integrity.
- **Standardization:** For clustering tasks, features were normalized using `StandardScaler` to ensure equal weighting across variables with different magnitudes (e.g., GDP vs. Life Expectancy).

C. Analytical Techniques

We employed several data mining techniques:

- 1) **Exploratory Data Analysis (EDA):** Correlation heatmaps and scatter plots to visualize relationships.
- 2) **Statistical Analysis:** Pearson correlation coefficients to quantify the strength of linear relationships.
- 3) **Unsupervised Learning:** K-Means clustering to categorize countries based on development indicators, using the Elbow Method to determine optimal clusters (k).
- 4) **Dimensionality Reduction:** PCA was used to visualize high-dimensional clusters in 2D space.

III. RESULTS

A. Life Expectancy Drivers

Correlation analysis identified the strongest determinants of life expectancy. As shown in Table I, Infant Mortality and Schooling Years have the most significant impact.

The moderate positive correlation with GDP (0.43) suggests that while economic wealth contributes to longevity, access to education and basic neonatal care (reflected in infant mortality) are more direct determinants.

TABLE I
PEARSON CORRELATION WITH LIFE EXPECTANCY

Feature	Correlation Coefficient (r)
Infant Mortality	-0.838
Adult Mortality	-0.696
Schooling Years	+0.714
GDP	+0.431

B. Vaccination and Infant Mortality

We analyzed the relationship between Measles vaccination coverage and Infant Mortality Rates (IMR) from 2010 to 2023.

- **Correlation:** A strong negative correlation ($r = -0.617, p < 0.001$) was observed.
- **Impact:** Statistical modeling suggests that every 10% increase in vaccination coverage saves approximately 4-5 infants per 1,000 births.
- **Regional Disparities:** Africa faces an IMR 18 times higher than Europe, largely correlated with a 28% deficit in vaccination coverage.

Specific country case studies highlight this trend:

- **India:** $\approx 90\%$ coverage $\rightarrow$ IMR ≈ 25 .
- **Nigeria:** $\approx 62\%$ coverage $\rightarrow$ IMR ≈ 70 .
- **San Marino:** $\approx 98\%$ coverage $\rightarrow$ IMR ≈ 1.5 .

C. Income Group Comparison

Health indicators vary drastically by income classification.

- **Mortality:** Adult mortality is highest in Low-Income groups (267 per 1000) compared to High-Income groups (120 per 1000).
- **HIV/AIDS:** Prevalence is highest in Low-Income regions (3.99%) versus High-Income regions (0.80%).
- **Spending Paradox:** Low-income nations spend a comparable percentage of GDP on health (6.02%) to high-income nations (6.43%), yet outcomes differ vastly, indicating that system efficiency and infrastructure are as important as spending volume.

D. Disease Trends: The Epidemiological Transition

Our analysis of mortality causes from 1990 to 2020 reveals a global shift:

- **Infectious Diseases:** Deaths from Malaria, TB, and HIV have declined significantly.
- **Chronic Diseases:** Mortality from Cardiovascular diseases, Neoplasms (Cancer), and Diabetes is rising.
- **Burden Shift:** As of 2020, chronic diseases account for approximately 77.5% of the analyzed disease burden, while infectious diseases account for 22.5%.

E. Clustering Analysis

Using K-Means clustering on standardized health and economic features, the Elbow Method indicated an optimal $k = 3$. PCA visualization confirmed distinct groupings:

- 1) **Cluster 0 (Underdeveloped):** High mortality, low schooling, low GDP.

- 2) **Cluster 1 (Developing):** Transitional economies, improving health metrics.
- 3) **Cluster 2 (Developed):** High life expectancy, high GDP, low disease burden.

Year-wise analysis of "Developing" nations showed a gradual migration of countries from lower-performing clusters to higher-performing ones over the last two decades.

IV. DISCUSSION

The data confirms that socio-economic development is the primary determinant of health. The strong inverse relationship between vaccination and mortality underscores the critical role of immunization programs. However, a significant "Health Gap" persists. While the gap in life expectancy between developed and developing nations has narrowed (from ≈ 13 years in 2000 to ≈ 11 years in 2015), it remains substantial.

Furthermore, the rise in chronic diseases presents a new challenge. Developing nations now face a "double burden" of disease: battling residual infectious diseases while simultaneously managing rising rates of non-communicable diseases due to urbanization and lifestyle changes.

V. CONCLUSION

This project successfully utilized data mining to evaluate global public health indicators. We conclude that:

- 1) Education and child survival are the strongest predictors of national life expectancy.
- 2) Vaccination programs have a statistically significant, high-impact role in reducing infant mortality.
- 3) The world is undergoing an epidemiological transition where chronic diseases are replacing infectious diseases as the leading causes of death.
- 4) Policy efforts must focus on closing the efficiency gap in health spending for low-income nations and addressing the rising tide of chronic illnesses.

APPENDIX

A. Team Contributions

- **Uday Kumar Choudhary:** Insights on Life Expectancy Drivers, correlation analysis.
- **Ayush Kumar Pandey:** Motivation, Vaccination impact analysis, Continent trends.
- **Nakul Sharma:** Data cleaning, Preprocessing, K-Means Clustering, PCA visualization.
- **Anuj Chhajed:** Correlation heatmaps, distribution plots, status-based comparisons.

B. Challenges Faced

The team encountered challenges regarding data consistency across international sources (WHO vs World Bank definitions) and handling missing values in historical records. These were mitigated by using robust imputation strategies and focusing on aggregated summary reports.

The complete source code, datasets, and analysis notebooks for this project are available at:

[https://github.com/Uday-Choudhary/
Global-Public-Health-Indicators.git](https://github.com/Uday-Choudhary/Global-Public-Health-Indicators.git)